immediate steps to move towards the centre. Unfortunately somebody has to be on the edge, but as far as each individual is concerned it is not going to be him! There will be a ceaseless migration in from the edges of an aggregation towards the centre. If the herd was previously loose and straggling, it will soon become tightly bunched as a result of the inward migration. Even if we start our model with no tendency towards aggregation at all, and the prey animals start by being randomly dispersed, the selfish urge of each individual will be to reduce his domain of danger by trying to position himself in a gap between other individuals. This will quickly lead to the formation of aggregations which will become ever more densely bunched.

Obviously, in real life the bunching tendency will be limited by opposing pressures: otherwise all individuals would collapse in a writhing heap! But still, the model is interesting as it shows us that even very simple assumptions can predict aggregation. Other, more elaborate models have been proposed. The fact that they are more realistic does not detract from the value of the simpler Hamilton model in helping us to think about the problem of animal aggregation.

The selfish-herd model in itself has no place for cooperative interactions. There is no altruism here, only selfish exploitation by each individual of every other individual. But in real life there are cases where individuals seem to take active steps to preserve fellow members of the group from predators. Bird alarm calls spring to mind. These certainly function as alarm signals in that they cause individuals who hear them to take immediate evasive action. There is no suggestion that the caller is ‘trying to draw the predator’s fire’ away from his colleagues. He is simply informing them of the predator’s existence—warning them. Nevertheless the act of calling seems, at least at first sight, to be altruistic, because it has the *effect* of calling the predator’s attention to the caller. We can infer this indirectly from a fact which was noticed by P. R. Marler. The physical characteristics of the calls seem to be ideally shaped to be difficult to locate. If an acoustic engineer were asked to design a sound that a predator would find it hard to approach, he would produce something very like the real alarm calls of many small song-birds. Now in nature this shaping of the calls must have been produced by natural selection, and we know what that means. It means that large numbers of individuals have died because their alarm calls were not quite perfect. Therefore there seems to be danger attached to giving